Machine Learning (2026) 115:243
https://doi.org/10.1007/s10994-026-07172-1

KARAMEL: Knowledge-Aware Ranking for Multimodal Entity Linking in the Arts via Heterogeneous Graphs

Raffaele Scaringi[1] · **Alejandro Sierra-Múnera**[2] · **Gennaro Vessio**[1] · **Giovanna Castellano**[1] · **Ralf Krestel**[3,4]

Received: 20 February 2026 / Revised: 8 September 2026 / Accepted: 16 September 2026
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Abstract
Discovering and disambiguating entities from multimodal and heterogeneous data is a key challenge in discovery science, where relevant knowledge is often implicit and distributed across modalities. We present KARAMEL, a framework for multimodal entity linking (MEL)—the task of aligning visual and textual mentions to entities in a knowledge base—in the cultural heritage domain. Unlike prior vision–language models that primarily rely on surface-level multimodal alignment, our approach incorporates sociohistorical and contextual knowledge through a knowledge-aware architecture that fuses a vision–language encoder with a heterogeneous graph neural network. This design enables contextual reasoning over complex relational structures and supports robust entity disambiguation under ambiguous or underspecified evidence. To benchmark multimodal discovery in this setting, we introduce $\mathcal{WikiMuSA}$ ($\mathcal{W}$ikidata-based $\mathcal{M}$ultimodal $\mathcal{S}$emantic data for $\mathcal{A}$rt), a new dataset that links artworks to Wikidata entities via images, textual descriptions, and structured contextual knowledge. Experimental results on both $\mathcal{WikiMuSA}$ and MELArt show that KARAMEL surpasses strong baselines, particularly on implicit or abstract mentions, while ablation studies highlight the critical role of domain-specific, structured context in learning expressive representations. These results suggest that KARAMEL has potential for other scientific applications where knowledge is critical for better entity linking. Both the dataset and the code are publicly available on Zenodo (https://zenodo.org/records/23019880) and GitHub (https://github.com/cilabuniba/karamel).

Keywords Multimodal entity linking · Knowledge discovery · Heterogeneous graph learning · Knowledge graphs · Cultural heritage

Raffaele Scaringi and Alejandro Sierra-Múnera contributed equally to this work.

Editors: Gianvito Pio, Jurica Levatić, Nikola Simidjievski.

Extended author information available on the last page of the article

1 Introduction

In recent years, computer vision and natural language processing have made significant strides, primarily driven by deep learning and large annotated datasets. In many real-world settings, however, the challenge is not only to process visual and textual signals but to *discover and disambiguate semantic entities* from multimodal and heterogeneous data, where relevant knowledge is often implicit and distributed across modalities. Vision and text transformers (Dosovitskiy et al., 2020; Vaswani, 2017) have achieved impressive results in various vision–language tasks, including image–text matching (Jiang et al., 2021), visual question answering (Ishmam et al., 2024), image captioning (Stefanini et al., 2022; Lu et al., 2022), and entity linking (Shen et al., 2021; Gjorgjevikj et al., 2025; Wang et al., 2024). However, these models often process visual and textual modalities in isolation, overlooking the broader contextual information that is crucial for knowledge discovery and robust semantic understanding.

This limitation becomes especially pronounced in the domain of cultural heritage, particularly in the analysis of artworks. Discovering connections among artistic entities requires going beyond digitized artwork images alone. Textual descriptions and structured knowledge bases encode rich social, historical, and political context that is essential for uncovering semantic relationships among entities and supporting reliable link discovery. Vision–text architectures often struggle to capture this contextual knowledge, which is essential for accurate interpretation. Furthermore, visual diversity in art—due to abstraction, stylistic variation, and subjective representation—complicates the use of visual features for tasks such as entity linking (Sierra-Múnera et al., 2024).

To address these challenges, recent approaches have explored integrating visual data with structured knowledge from knowledge graphs (KGs), achieving promising results in tasks such as artwork classification (Scaringi et al., 2025). In this context, *multimodal entity linking* (MEL)—the task of identifying and linking entities mentioned in text and depicted in images to a knowledge base—can be viewed as a knowledge discovery problem over complex, multimodal data. Some MEL models represent context by verbalizing entity attributes derived from KGs (Luo et al., 2023), but this approach captures only a partial view of an entity's background. General-purpose KGs such as Wikidata contain extensive cultural heritage information (Baroncini et al., 2023); however, directly applying graph embedding models to such large graphs raises scalability and computational challenges. Recent work in knowledge extraction (Jarnac et al., 2023) has therefore focused on identifying relevant subgraphs, though aggressive pruning may introduce inconsistency or data drift, ultimately limiting the reliability of contextual information.

Addressing these challenges requires models that go beyond surface-level multimodal alignment and are able to reason over contextual knowledge. In this work, we propose KARAMEL, a knowledge-aware architecture for multimodal entity linking that explicitly integrates visual and textual information with a third modality: structured contextual knowledge encoded in a heterogeneous knowledge graph. By combining a vision–language backbone with an inductive heterogeneous graph neural network, KARAMEL enables contextual reasoning over relational structures and supports more expressive and accurate entity discovery. We extract a focused artistic subgraph from Wikidata, forming $\mathcal{WikiMuSA}$, a lightweight KG tailored to the domain of art, and demonstrate that domain-specific contextual knowledge significantly improves disambiguation beyond vision–text inputs alone.

KARAMEL is specifically designed to address the unique challenges of MEL in the artistic domain. As shown in Fig. 1b, the task involves linking depicted entities from an artwork and its textual description without assuming explicit mentions in the text. For instance, while *Arch of Nero* is mentioned, the link with the entity *ruins* must be inferred from visual evidence combined with contextual cues such as "*ancient Roman architecture*". This contrasts with standard MEL settings (Fig. 1a), where entities are explicitly named and easily extracted. Our dataset, therefore, introduces a more challenging and realistic discovery setting compared to existing benchmarks such as Wikidata-MEL (Wang et al., 2022) or MELArt (Sierra-Múnera et al., 2024).

A second challenge in this domain is the visual complexity of artworks: depicted entities often occupy only small regions of an image, while variations in artistic style and brushwork further hinder reliable visual matching. These challenges underscore the need to integrate structured contextual knowledge, enabling more robust entity discovery and disambiguation beyond the capabilities of vision–language models operating in isolation.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the proposed dataset. Section 4 introduces our framework. Section 5 presents the experimental evaluation. Finally, Sect. 6 concludes the paper and outlines future directions.

2 Related Work

Entity linking (EL) is the task of mapping textual mentions to corresponding entities within a knowledge base. It can be viewed as a knowledge discovery problem in which latent semantic references must be inferred from incomplete or ambiguous evidence (Dost et al., 2020; Shen et al., 2023). Early methods used word embeddings, such as Word2Vec (Church, 2017), in combination with convolutional neural networks (Li et al., 2021) to link mentions to entities (Cao et al., 2017). Recognizing the limitations of pure text-based representations, Gupta et al. (2017) proposed incorporating coarse- and fine-grained features to construct more accurate entity representations.

With the rise of transformer-based architectures (Vaswani, 2017; Zhang and Shafiq, 2024), several works have adapted them to enhance EL. Peters et al. (2019) introduced a knowledge-enhanced BERT model that jointly encodes mentions and entities using a cross-attention module to produce context-aware representations. Wu et al. (2019) proposed BLINK, a two-stage approach in which BERT encodes mentions and a separate transformer

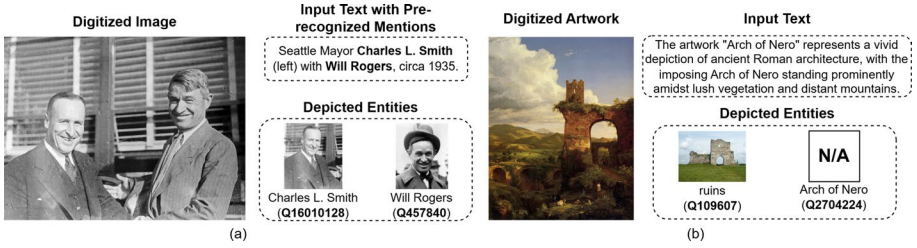


Fig. 1 Comparison between the standard multimodal entity linking setting (**a**), where entity mentions are explicitly highlighted in the text, and our proposed setting (**b**), where mentions must be inferred from raw Wikipedia articles without predefined annotations

ranks candidate entities. Mulang et al. (2020) propose a technique to include lexicalized KG triples as part of the context of a bi-directional transformer encoder, to improve the disambiguation between candidates. However, the number of triples that can be encoded for each candidate entity is limited by the context length. Ayoola et al. (2022) also leverages a KG by assigning a score to the combination of candidate entities based on their existing relations in the KG. A prerequisite to this approach is the recognition of multiple named entities in the text. Li et al. (2023) proposed to inject entity labels as synsets into BERT using an entity-aware adapter. More recently, Zhang et al. (2024) addressed temporal entity linking by integrating temporal graph data. However, their approach primarily focused on structural and textual signals, without leveraging node-level feature engineering to enhance semantic expressiveness further.

Although textual signals have proven effective in many EL scenarios, visual information can be critical in disambiguation, especially when entities or mentions are visually grounded. Multimodal entity linking extends traditional EL by incorporating visual features along with textual features (Bielefeld et al., 2024; Gan et al., 2021). Early work by Moon et al. (2018) showed that combining visual and textual information can alleviate polysemy in social media analysis (Balaji et al., 2021). Later studies explored deeper modality interactions: for example, Wang et al. (2022) used cross-attention mechanisms to fuse visual and textual signals, while Luo et al. (2023) introduced a multi-grained matching module to align multimodal entity-mention representations more effectively. Song et al. (2024) introduced a dual-way enhanced framework that learns interactions between the modalities based on the cross-attention mechanism, facial features, and adjective-noun phrases. More recent MEL works used large language models (LLMs) and Multimodal large language models (MLLMs) as integral components for solving MEL. Liu et al. (1909) proposed augmenting the mention and entity descriptions using an MLLM and an LLM respectively, before encoding them in a common vector space. They also fine-tuned an LLM to deliver a final answer feeding a prompt that included mention information and the top-k candidates in the vector space. Shi et al. (2024) employed in-context learning by incorporating example instances into prompts provided to a large language model, and complemented this approach with constrained decoding that limits generated outputs to entity names contained within a knowledge base.. Kim et al. (2025) decomposed the MEL task into three stages. First, a MLLM extracts a knowledge graph from the mention text, centered on the referenced entity. Next, the extracted graph, together with the image and text, is embedded into a shared vector space using CLIP (Radford et al., 2021), enabling retrieval of the top-k candidate entities. Finally, an LLM performs the entity linking task by selecting the correct entity from the retrieved candidates based on an MEL prompt.

However, MEL becomes particularly challenging in domains where context plays a central role. The cultural heritage domain exemplifies this, as it involves complex, often implicit, contextual relationships. In recent years, fine arts have become a growing focus within multimodal AI research, leading to advances in tasks such as artwork classification (Scaringi et al., 2025), style transfer (Guo et al., 2024; Dong et al., 2025), inpainting (Fanelli et al., 2024), and guided artwork generation (Tang et al., 2025). These efforts span predictive and generative settings and have spurred the creation of domain-specific datasets.

Among them, $\mathcal{ArtGraph}$ (Castellano et al., 2022) is a large-scale KG constructed from sources such as WikiArt (Castellano and Vessio, 2021), DBpedia, and Artemis (Achlioptas

et al., 2021), representing artworks, artists, and emotions evoked by artworks. Although rich in general contextual information, $\mathcal{ArtGraph}$ lacks a detailed taxonomy for the depicted entities, limiting its application in fine-grained MEL tasks. Similarly, Artpedia (Stefanini et al., 2019) classifies sentences from Wikipedia painting articles into visual and contextual categories, providing a textual resource for multimodal analysis. Based on this, MELArt (Sierra-Múnera et al., 2024) connects the iconographic entities mentioned in Artpedia's visual sentences to KG entries, forming an automatically annotated MEL dataset. However, MELArt assumes that the depicted entities are explicitly mentioned in the text and does not aim to cover all entities depicted in an artwork, thereby limiting its generalizability.

To address these gaps, we propose a knowledge-aware perspective on multimodal entity linking that explicitly leverages structured relational context for discovery and disambiguation. Within this framework, we introduce $\mathcal{WikiMuSA}$, a curated subgraph extracted from Wikidata that focuses on domain-relevant relationships in the artistic domain, balancing information richness and scalability. Building on this resource, we present KARAMEL, a multimodal framework that integrates images, text, and heterogeneous graph-based knowledge to learn expressive representations of depicted entities.

3 $\mathcal{W}$iki$\mathcal{M}$u$\mathcal{SA}$

Understanding artworks and the entities they depict requires discovering semantic relationships across multimodal, heterogeneous sources, where relevant context is often implicit rather than explicitly observed. Visual and textual cues alone are frequently insufficient, as historical, social, political, or religious background plays a central role in interpretation. For example, knowing that another mentored an artist or that a depicted figure held a significant public role or personal relationship can enrich interpretation. This contextual layer can be modeled through both structured metadata and unstructured descriptions. A comprehensive knowledge graph allows such connections to be explicitly represented, capturing interdependencies and domain semantics.

To this end, we introduce $\mathcal{WikiMuSA}$, a dataset in the artistic domain comprising visual, textual, and contextual information relevant to cultural heritage. Figure 2 illustrates its core schema, focusing on the relationships of entities and their domain-specific properties.

3.1 Data Collection and Pre-processing

We constructed $\mathcal{WikiMuSA}$ from artworks in Wikidata that meet the following criteria: (*i*) a linked *Wikipedia article*; (*ii*) at least one *depicts* relation to another Wikidata entity; and (*iii*) an associated *image*. Wikipedia articles were retrieved using the Huggingface dataset *wikimedia/wikipedia*,[1] based on the 2023-11-01 dump. Since articles span multiple languages, we used the NLLB-200 (3.3B) model (Costa-jussà et al., 2022) to translate non-English texts into English, thereby allowing us to maximize the number of artworks included in the dataset regardless of the original article language. We then used LLaMA 3 (8B) (Grattafiori et al., 2024) to generate concise, depiction-focused summaries. This summarization step was introduced to produce compact textual descriptions that can be effectively processed by lightweight text encoders, avoiding the need to rely on LLMs for textual feature extraction.

[1] https://huggingface.co/datasets/wikimedia/wikipedia

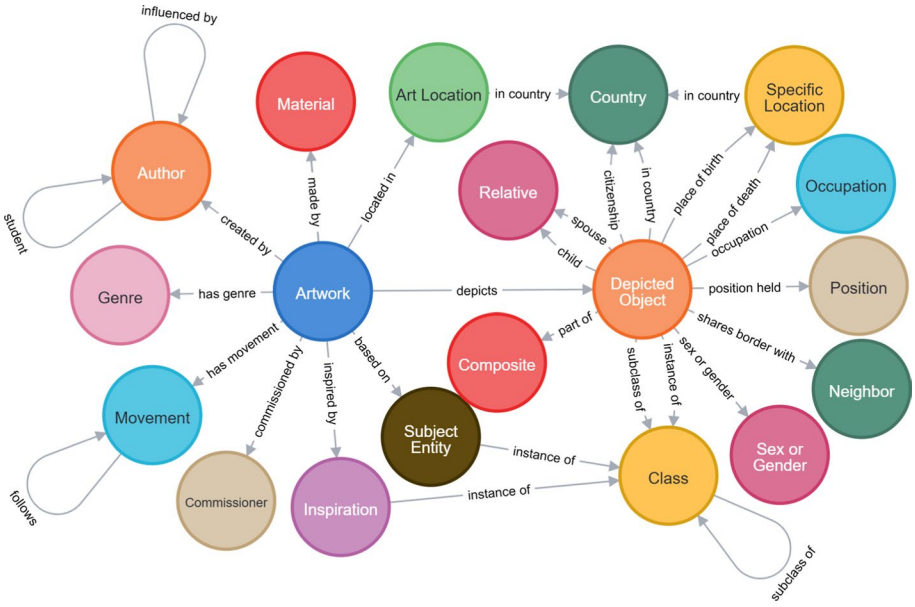


Fig. 2 Logical schema of *WikiMuSA*. Each artwork is associated with metadata, including author, genre, and movement. Depicted entities are linked through enriched, contextual relationships that enhance expressiveness and graph connectivity

This design choice enables a more faithful evaluation of a model's ability to generalize to unseen artworks, rather than relying on the extensive world knowledge encoded in LLMs, which are likely to have encountered many of the underlying Wikipedia articles during pretraining. In addition, this summarization step was designed to reduce variance and noise across heterogeneous sources while preserving depiction-relevant content, and was applied as a standalone preprocessing step, ensuring that all experimented models were evaluated under identical conditions. The distribution of original languages and the length of the articles before and after summarization are discussed in detail in Sect. 7.

3.2 Graph Structure and Relations

Each artwork is represented as a node connected to multiple nodes of contextual information, including author, genre, artistic movement, gallery, and depicted entities. These depicted entities can be people, places, abstract concepts, and more. For each category, we selected a curated set of relevant properties: *people* (occupation, gender, relatives); *places* (neighboring cities and location hierarchy, e.g., a city within a country), and *galleries* linked to the city and hosting relationships. Each node in *WikiMuSA* could include visual and textual features. Visual data, if available, is sourced from the Wikidata image property (linked to Wikimedia Commons), while textual information is created by concatenating the entity's English label and description.

3.3 Final Dataset

The resulting dataset is a multimodal resource (vision, language, context) comprising 13, 482 unique artworks and 12, 177 unique depicted entities. Each artwork is associated with an image and a summarized Wikipedia article, while each depicted entity is associated with an optional image, a description, and a structured context from $\mathcal{W}$ *iki* $\mathcal{M}$ *u* $\mathcal{SA}$. Hence, artworks are modeled using image-text pairs, with the depicted entities represented by visual features, textual descriptions, and graph-derived contextual embeddings. This setup provides a rich testbed for multimodal entity linking in the arts and, more broadly, for studying knowledge-aware discovery from multimodal and heterogeneous data in domains where contextual information plays a central role.

Table 1 compares *WikiMuSA* with other MEL datasets. The main difference with all the existing datasets is the inclusion of implicit mentions (e.g., "ruins" in Fig. 1). The other difference is the introduction of a contextual graph that describes the entities. We note that Wikidata-MEL and MELArt do not provide a KG as part of the dataset; however, the entities are aligned with Wikidata by their identifiers, facilitating the construction of a contextual graph as we explored in our experiments with MELArt. Finally, compared with MELArt, *WikiMuSA* covers more artworks and, by including more languages in the Wikipedia articles, broadens the spectrum of artistic representation beyond the English Wikipedia.

4 Proposed Method

We propose KARAMEL, a context-aware framework for multimodal entity linking that explicitly addresses entity discovery under ambiguous and incomplete evidence. Building on recent advances in vision–language models, KARAMEL extends vision–text alignment by incorporating structured contextual knowledge from a heterogeneous knowledge graph through a dedicated graph encoder.

4.1 Problem Formulation

The goal of MEL is to link a set of multimodal *mentions* to corresponding *entities* using multiple information sources, such as visual and textual signals. Formally, we define the

Table 1 Comparison of state-of-the-art multimodal entity linking (MEL) datasets and the proposed *WikiMuSA* dataset across domain, scale, mention types, and contextual graph availability.

Dataset	Domain	# Mentions	# Candidate Entities	Mention Type		Contextual Graph
				Explicit	Implicit	
Wikidata-MEL (Wang et al., 2022)	General	22,136	17,391	✓	✗	✗
Richpedia-MEL (Wang et al., 2020)	General	17,724	17,805	✓	✗	✓
WikiDiverse (Wang et al., 2022)	News	7,824	7,764	✓	✗	✗
MELArt (Sierra-Múnera et al., 2024)	Art	1,616	53,901	✓	✗	✗
WikiMuSA (*ours*)	Art	13,482	17,177	✓	✓	✓

Mention Type indicates whether the entity is explicitly stated in the textual mention or not, while Contextual Graph indicates whether a dataset provides an associated graph counterpart.

mention set as $\mathcal{M} = \{\mathrm{M}_i\}_{i=1}^{N}$, where each mention M_i consists of a visual representation M_{vi} and a textual description M_{ti}, that is, $\mathrm{M}_i = (M_{vi}, M_{ti})$. Similarly, the set of candidate entities is $\mathcal{E} = \{\mathrm{E}_j\}_{j=1}^{P}$, with each entity $\mathrm{E}_j = (E_{vj}, E_{tj})$ including an image and a textual description.

We also introduce a heterogeneous knowledge graph $\mathcal{G} = (\mathcal{V}, \mathcal{L})$, where $\mathcal{V} = \{v_1, \ldots, v_{|\mathcal{V}|}\}$ is the set of nodes and $\mathcal{L} \subseteq (\mathcal{V} \times \mathcal{V}) = \{l_1, \ldots, l_{|\mathcal{L}|}\}$ is the set of edges. Each node $v \in \mathcal{V}$ is associated with a type via a function $\mu : \mathcal{V} \to \mathcal{T}_{\mathcal{V}}$, where $t_v = \mu(v)$ denotes its node type (e.g., *Artwork*, *Depicted Object*, *Artist*). Additionally, we define the support of each node type t as the set of nodes associated with t. In general, $\mathcal{V}_t = \{v \in \mathcal{V} \,|\mu(v) = t\}$. Each edge $l \in \mathcal{L}$ is similarly typed through a function $\rho : \mathcal{L} \to \mathcal{T}_{\mathcal{L}}$, where $t_l = \rho(l)$ defines its semantic relation (e.g., *depicts*, *located in*, *instance of*, etc.). We assume that for every entity $\mathrm{E}_j \in \mathcal{E}$ there exists a corresponding node $v_j \in \mathcal{V}$ of type *Depicted Object*. Given an input mention M_i, the goal of MEL is to correctly identify the ground-truth entity $\mathrm{E}_j \in \mathcal{E}$ by maximizing its probability among all candidates. This task can be formalized as the maximization of the log-likelihood over a training set $\mathcal{D}_{\text{train}}$:

$$\theta^* = \arg\max_{\theta} \sum_{(\mathrm{M}_i, \mathrm{E}_j) \in \mathcal{D}_{\text{train}}} \log p_\theta(\mathrm{E}_j \mid \mathrm{M}_i, \mathcal{E}). \tag{1}$$

To incorporate contextual knowledge from the KG, we extend the objective to condition on $\mathcal{G}$ as well:

$$\theta^* = \arg\max_{\theta} \sum_{(\mathrm{M}_i, \mathrm{E}_j) \in \mathcal{D}_{\text{train}}} \log p_\theta(\mathrm{E}_j \mid \mathrm{M}_i, \mathcal{E}, \mathcal{G}). \tag{2}$$

This formulation encourages the model to leverage all three modalities—vision, text, and context—during entity ranking.

4.2 Model Architecture

KARAMEL jointly processes an artwork mention M_i, a candidate entity E_j, and a knowledge graph $\mathcal{G}$ using three main architectural components: an image-text cross-encoder, a heterogeneous graph neural network (GNN) encoder, and a cross-modal ranker, as illustrated in Fig. 3. The image-text cross-encoder extracts expressive features from both images and textual descriptions associated with artwork mentions and depicted entities, using a CLIP architecture (Radford et al., 2021) pre-trained on the LAION-400M dataset (Schuhmann et al., 2021), whereas the GNN encoder exploits graph convolutions and the structured semantics in the input KG to extract context-aware information for all depicted entities, improving their expressiveness.

4.2.1 Feature Encoding

4.2.1.1 Visual Feature Extraction The CLIP vision encoder employs a vision transformer (ViT) (Dosovitskiy et al., 2020) as its backbone. Given an RGB image I with dimensions $C \times H \times W$—either the artwork image M_{vi} or the entity image E_{vj}—ViT encodes it into a hidden representation $\mathrm{V} \in \mathbb{R}^{(n+1)\times d}$: $\mathrm{V} = \left[\mathrm{v}^0_{[CLS]}, \mathrm{v}^1, \ldots, \mathrm{v}^n\right]$. We extract two types of visual

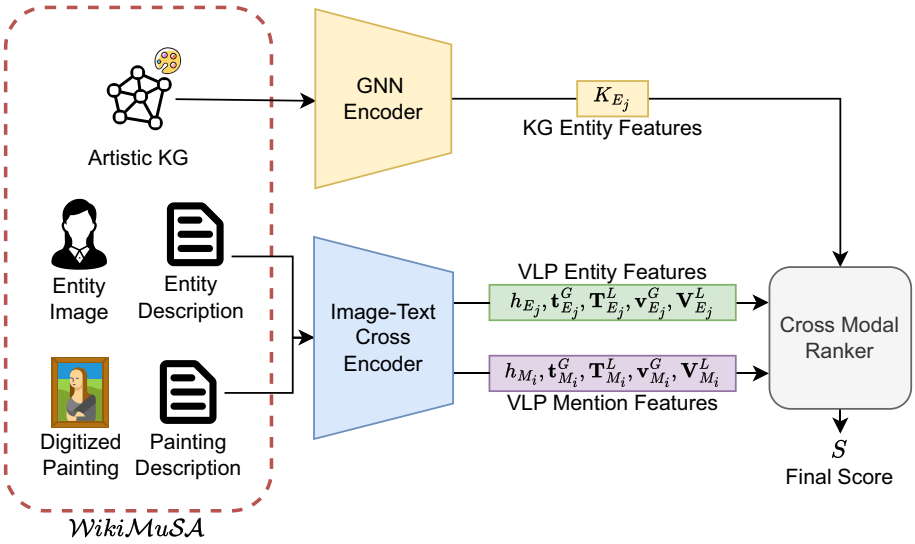


Fig. 3 Architecture of KARAMEL. The image-text cross-encoder and the GNN encoder are jointly trained to extract complementary information from artworks and depicted entities. The cross-modal ranker is then trained to rank candidate entities effectively for a given input mention.

features: the global visual representation, corresponding to the [CLS] token, $\mathrm{v}_G \in \mathbb{R}^d$; and the local visual representation, given by all token embeddings, $\mathrm{V}_L \in \mathbb{R}^{(n+1)\times d}$. For artwork mentions, these are denoted as $\mathrm{v}^G_{M_i}$ and $\mathrm{V}^L_{M_i}$, respectively. For entities, we similarly define $\mathrm{v}^G_{E_j}$ and $\mathrm{V}^L_{E_j}$. When entities lack visual information, we apply zero-padding as standard practice.

4.2.1.2 Textual Feature Extraction For textual encoding, we use the CLIP text encoder, which is a transformer-based architecture analogous to BERT (Devlin et al., 2019). Given an input text sequence S consisting of l_s tokens, the text encoder produces a contextualized embedding matrix $\mathrm{T} \in \mathbb{R}^{(l_s+1)\times d}$: $\mathrm{T} = \left[\mathrm{t}^0_{[CLS]}, \mathrm{t}^1, \ldots, \mathrm{t}^{l_s}\right]$. As with visual features, we extract the global textual representation of the [CLS] token, $\mathrm{t}_G \in \mathbb{R}^d$, and the local textual representation of the entire token sequence, $\mathrm{T}_L \in \mathbb{R}^{(l_s+1)\times d}$. We denote the textual representations of the artwork mentions as $\mathrm{t}^G_{M_i}$ and $\mathrm{T}^L_{M_i}$, and those of entities as $\mathrm{t}^G_{E_j}$ and $\mathrm{T}^L_{E_j}$.

4.2.1.3 Graph Feature Extraction The graph encoder employs a heterogeneous GNN to extract contextual embeddings from $\mathcal{G}$, focusing on nodes representing depicted entities. Let $t =$ "depicted entity" be the target node type. The consequent knowledge graph embedding matrix encompassing all candidate depicted entities can be defined as:

$$\mathrm{K} = \left[\mathrm{k}_{E_j} = \mathrm{x}^{(L)}_{v_j} \,|v_j \in \mathcal{V}_t\right], \tag{3}$$

where each depicted entity $E_j \in \mathcal{E}$ is encoded as $\mathrm{k}_{E_j} \in \mathbb{R}^d$, $\mathrm{x}_{v_j}^{(L)}$ is the learned embedding of node v_j at the last GNN layer L, and $\mathcal{V}_t$ is the set of nodes having type t.

In general, each GNN layer updates node representations by aggregating information from neighboring nodes through permutation-invariant operations and transforming it via learnable functions. This message-passing mechanism enables nodes to incorporate contextual information from their local graph neighborhood. Node neighborhoods are defined by the graph connectivity, and the aggregation process is applied consistently across all nodes. Initial node features are derived from node-specific metadata; in our implementation, we encode each Wikidata entity by concatenating its label and description, then project the resulting text into a dense embedding space using the same CLIP text encoder.

4.2.2 Cross-Modal Ranker

To compute a ranking score for each mention-entity pair, KARAMEL combines multiple modalities to explore intra- and inter-modal interactions among visual, textual, and knowledge graph information. Specifically, we define four components that contribute to the overall score: a vision-based score, a text-based score, a vision-text cross-modal score, and a contextual knowledge score.

4.2.2.1 Vision-Based Score Visual information is fundamental in MEL, particularly in the art domain, where artworks are primarily expressed through visual characteristics: from low-level features like brushstrokes to high-level representations such as depicted objects and symbolic motifs. We adopt the dual-gated mechanism introduced in MIMIC (Luo et al., 2023) to model visual interaction, which computes the cross-attention between mention and entity features. Formally, the vision-based score is computed as:

$$S_V = \mathcal{U}_V(\mathrm{M}_i, \mathrm{E}_j) = (S_V^{E2M} + S_V^{M2E})/2,\ S_V \in \mathbb{R} \tag{4}$$

where S_V^{E2M} and S_V^{M2E} are obtained using cross-visual attention between mention and entity representations, having $S_V^{A2B} = \Phi_{A2B}(\mathrm{v}_A^G, \mathrm{v}_B^G, \mathrm{V}_B^L)$, where $\Phi_{A2B}(\cdot,\cdot,\cdot)$ is the scoring attention module, and v_A^G and V_B^L are the global and local vision representations defined in Sect. 4.2.1.

4.2.2.2 Text-Based Score Text remains a core modality in MEL. Following (Luo et al., 2023), we compute a text-based score using the text-based global-local interaction unit (TGLU), which models both global and local textual dependencies:

$$S_T = \mathcal{U}_T(\mathrm{M}_i, \mathrm{E}_j) = (S_T^{G2G} + S_T^{G2L})/2,\ S_T \in \mathbb{R} \tag{5}$$

where $S_T^{G2G} \in \mathbb{R}^d$ is the global-to-global score, formalized as $S_T^{G2G} = \mathrm{t}_{E_j}^G \cdot \mathrm{t}_{M_i}^G$, and the global-to-local score is a gated cross-attention mechanism $S_T^{G2L} = \mathrm{t}_{E_j}^G \cdot ATTN(\mathrm{T}_{E_j}^L, \mathrm{T}_{M_i}^L, \mathrm{T}_{M_i}^L),\ S_T^{G2L} \in \mathbb{R}^d$.

4.2.2.3 Vision-Text Cross-Modal Score To jointly model visual and textual information, we compute a vision-text cross-modal score:

$$S_{VT} = \mathcal{U}_{VT}(\mathrm{M}_i, \mathrm{E}_j) = \mathrm{h}_{E_j}^{T} \mathrm{h}_{M_i},\ S_{VT} \in \mathbb{R} \tag{6}$$

where $\mathrm{h}_{E_j} \in \mathbb{R}^d$ and $\mathrm{h}_{M_i} \in \mathbb{R}^d$ are cross-modal representations regarding the mention M_i and the entity E_j, respectively. More precisely, those cross-attended representations are obtained as $\mathrm{h}_{M_i} = CMFU(\mathrm{V}_{M_i}^{L}, \mathrm{T}_{M_i}^{L}, \mathrm{v}_{M_i}^{G}, \mathrm{t}_{M_i}^{G})$, or for an entity $E_j \in \mathcal{E}$ $\mathrm{h}_{E_j} = CMFU(\mathrm{V}_{E_j}^{L}, \mathrm{T}_{E_j}^{L}, \mathrm{v}_{E_j}^{G}, \mathrm{t}_{E_j}^{G})$, where *CMFU* (Luo et al., 2023) embodies a multi-grained cross-attention strategy for gathering a comprehensive representation of the mention or the entity.

4.2.2.4 Contextual Knowledge Score Finally, we integrate contextual information from the knowledge graph by computing an alignment between the mention's cross-modal representation and the entity's KG embedding:

$$S_C = \mathcal{U}_C(\mathrm{M}_i, \mathrm{E}_j) = \mathrm{k}_{E_j}^{T} \mathrm{h}_{M_i},\ S_C \in \mathbb{R} \tag{7}$$

where $\mathrm{k}_{E_j} \in \mathbb{R}^d$ is the knowledge graph embedding for the entity $E_j \in \mathcal{E}$ obtained using Eq. 3, and $h_{M_i} \in \mathbb{R}^d$ is the cross-attended vision-text mention representation.

4.2.2.5 Final Ranking Score Combining all the components above, the final relevance score for a mention-entity pair is defined as the average of the four scores:

$$S = \mathcal{U}(\mathrm{M}_i, \mathrm{E}_j) = (S_V + S_T + S_{VT} + S_C)/4 \tag{8}$$

This ranking function enables KARAMEL to reason beyond surface features by incorporating structured contextual knowledge, capturing fine information such as historical, religious, and social factors. Unlike current frameworks, our knowledge-aware design facilitates discovery and disambiguation in multimodal settings by leveraging structured relational context. While focused on the artistic domain, the approach may also be applicable to other multimodal scenarios with structured contextual information.

5 Experiments

We conducted a comprehensive set of experiments on the proposed dataset to answer the following research questions:

RQ1 How does KARAMEL perform in ranking candidate entities compared to the baseline models on artistic datasets?

RQ2 Which characteristics of the data most influence the performance of KARAMEL?

RQ3 How does model performance vary under different feature encoding strategies?

We evaluated KARAMEL against several entity linking baselines, training all models on both $\mathcal{WikiMuSA}$ (see Sect. 3) and MELArt (Sierra-Múnera et al., 2024). During the evaluation, the candidate universe is the complete set of depicted entities. These are ranked for each artwork in the test set, and we report standard retrieval metrics: recall at rank k (R@K), mean average precision (MAP), and mean reciprocal rank (MRR). In our task, where each input mention may have multiple relevant entities, MAP captures the average precision across all ground-truth entities. At the same time, MRR focuses only on the first correctly ranked entity. For all metrics, higher values are better.

5.1 Experimental Setting

We randomly split each evaluated dataset into training, validation, and test sets using an 80/10/10 ratio, based on artworks. To prevent data leakage, all validation and test artworks were removed from the graph $\mathcal{G}$. This ensures that no validation or test artwork nodes are present in the graph, thereby eliminating any indirect influence on the entity embeddings' learning objective. Furthermore, all images (for mentions and entities) were resized to 224×224 pixels. For entities without associated images, we applied zero-padding. For visual and textual feature encoding (see Sect. 4.2.1), we used a pre-trained CLIP (base) model (Radford et al., 2021). For the graph encoder, we trained a heterogeneous GNN from scratch using GraphSAGE layers (Hamilton et al., 2017).

Regarding textual input for mentions, we diverged from the standard entity linking formulation, which typically formats mentions as:

$$M_{it} = [\text{CLS}]\text{m}_l[\text{SEP}]\text{m}[\text{SEP}]\text{m}_r, \tag{9}$$

that is, the mention surrounded by left and right contexts. Instead, we use the entire summarized article without predefining mention spans. This challenges the model to identify relevant text portions associated with each entity, rather than relying on annotated mentions, and, at the same time, allows models to use a single mention representation for multiple depicted entities.

Inspired by Luo et al. (2023), we employed a unit-consistent contrastive loss that accounts for all three modalities. Based on the composite scores defined in Eqs. 4–8, we first defined the main contrastive loss as:

$$\mathcal{L}_O = -\log \frac{\exp(\mathcal{U}(\text{M}_\text{i}, \text{E}_j))}{\sum_k \exp(\mathcal{U}(\text{M}_i, \text{E}_k))}, \tag{10}$$

where E_k denotes negative samples within the same batch. To promote disentangled representations and assess the contribution of individual modalities, we further computed auxiliary losses for each interaction component:

$$\mathcal{L}_X = -\log \frac{\exp(\mathcal{U}_X(\text{M}_\text{i}, \text{E}_j))}{\sum_k \exp(\mathcal{U}_X(\text{M}_i, \text{E}_k))}, X \in \{T, V, VT, C\}. \tag{11}$$

Here, X refers to the textual (T), visual (V), vision-text (VT), and contextual (C) interaction scores defined in Sect. 4.2.2. The final loss function is the sum of all components:

$$\mathcal{L} = \mathcal{L}_O + \mathcal{L}_V + \mathcal{L}_T + \mathcal{L}_{VT} + \mathcal{L}_C. \tag{12}$$

This promotes consistency across modalities and ensures robust mention-entity matching.

We trained the model for 150 epochs with a batch size of 64. We used the AdamW optimizer (Loshchilov & Hutter, 2017), a step-wise cosine decay scheduler (Loshchilov & Hutter, 2016), and a linear warm-up of with number of steps equal to the number of training batches. The maximum learning rate was set to $1e{-}5$. We applied early stopping based on MAP and terminated training if no improvement was observed for 10 consecutive validation epochs.

All experiments were repeated five times with different seeds. The experiments were conducted on a single NVIDIA A100 GPU (64 GB VRAM) with 512 GB RAM, ensuring sufficient computational resources for training on our large multimodal dataset.

5.2 Baselines

We compared KARAMEL against a wide number of representative baselines, covering text-based, vision-language pre-trained (VLP), multimodal, KG-based, and LLM/MLLM models. Specifically, we compared with:

- BERT (Devlin et al., 2019), implemented as a shared-parameter text bi-encoder that independently encodes mentions and entities into a common embedding space, with cosine similarity used as the matching score;
- CLIP (Radford et al., 2021), implemented as a shared-parameter multimodal bi-encoder with cosine similarity for retrieval, evaluated in both zero-shot and fine-tuned settings;
- ViLT (Kim et al., 2021), employed as a native vision-language cross-encoder that jointly encodes image–text pairs into a unified representation for mention–entity matching;
- MIMIC (Luo et al., 2023), a state-of-the-art multimodal model tailored for MEL that integrates multigrained analysis of vision and text;
- Mulang et al. (2020), a knowledge graph-enhanced entity linking model that incorporates lexicalized KG triples into the context of a bidirectional transformer encoder to enrich candidate entity representations and support disambiguation;
- UniMEL (Liu et al., 1909), an LLM- and MLLM-based entity linking framework that augments mention and candidate descriptions using generative models prior to vector space encoding, coupled with an LLM for candidate reranking;
- GEMEL (Shi et al., 2024), a generative multimodal entity linking framework that leverages frozen vision and language models with a lightweight cross-modal feature mapper to directly generate target entity names via in-context demonstrations;
- KGMEL (Kim et al., 2025), a knowledge-graph-based multimodal framework that extracts a knowledge graph from mention text via an MLLM, retrieves candidate entities in a shared multimodal space, and employs an LLM for final entity disambiguation.

This diverse set of baselines enables a comprehensive comparison across different modalities and architectural paradigms. Comparing against unimodal text models (BERT) and

multimodal architectures (CLIP, ViLT, MIMIC) allows us to isolate the impact of visual grounding and cross-modal interactions versus purely textual representations. Furthermore, including KG-based approaches (Mulang et al., KGMEL) and generative LLM/MLLM-based architectures (UniMEL, GEMEL) allows us to assess the effectiveness of our heterogeneous graph neural network formulation against alternative strategies for incorporating structured knowledge and foundation model reasoning for entity disambiguation.

Notably, multiple MEL models depend on the prior recognition of entities in the text to support the disambiguation and linking of the corresponding entities. In our experiments, we adapt Mulang et al., KGMEL, UniMEL, and GEMEL so that they don't depend on mention spans, thereby forcing them to recognize and rank entities using only the full text and the image. Furthermore, GEMEL does not produce rankings, but rather directly generates the entity description using constrained generation. We therefore only compute Recall@1 with a single output.

5.3 Experimental Results

To address the research questions and assess the impact of the key design choices in KARAMEL, we conducted a series of comparative experiments. We first evaluated performance using ranking metrics to compare KARAMEL against baseline methods, and then performed a detailed analysis of individual components, as well as studies on complexity and scalability. Finally, we investigated the effects of translation and summarization in the construction of $\mathcal{WikiMuSA}$.

5.3.1 Performance Comparison

To answer RQ1, we compared KARAMEL with the selected baselines in Table 2, benchmarking on both $\mathcal{WikiMuSA}$ and MELArt (Sierra-Múnera et al., 2024) datasets, which we adapted to our experimental setting by extracting a graph with the same ontology as $\mathcal{W}$ *iki* $\mathcal{M}$ *u* $\mathcal{SA}$ seen in Fig. 2, containing the artworks and depicted entities in MELArt. For $\mathcal{W}$ *iki* $\mathcal{M}$ *u* $\mathcal{SA}$ across all metrics, KARAMEL outperforms the baselines, with MIMIC being the closest competitor. This demonstrates that incorporating contextual information from a knowledge graph leads to more accurate and robust entity linking, particularly in complex domains such as cultural heritage.

Although being a unimodal baseline, BERT achieves surprisingly strong results, outperforming CLIP (in both zero-shot and fine-tuning settings) and ViLT despite using only textual information. This suggests that the modality gap (Liang et al., 2022) in vision-language pre-training may hinder the performance of contrastive models, such as CLIP and ViLT, in this setting. Moreover, BERT benefits from dense textual supervision, which is especially useful when entities are described in rich natural language, as in Wikipedia. Moving beyond unimodal and non-KG architectures, the KG-based approaches (Mulang et al. and KGMEL) demonstrate that incorporating structural information yields a clear advantage over traditional text and vision-language baselines on the $\mathcal{W}$ *iki* $\mathcal{M}$ *u* $\mathcal{SA}$ dataset, though they still fall short of KARAMEL. As for the generative LLM/MLLM-based models, i.e., UniMEL and GEMEL, they achieve competitive results but similarly remain behind our approach. Interestingly, although large language models have certainly been exposed to extensive cultural heritage data during their pretraining, they still struggle to correctly reason over complex

Table 2 Ranking metrics for the baselines and KARAMEL on both *WikiMuSA* and MELArt (Sierra-Múnera et al., 2024) datasets.

WikiMuSA						
Model	R@1	R@10	R@50	R@100	MAP	MRR
BERT (Devlin et al., 2019)	02.06 (± 1.11)	09.67 (± 2.81)	23.08 (± 4.10)	31.62 (± 3.18)	08.15 (± 2.56)	10.52 (± 2.90)
CLIP (Radford et al., 2021)	00.82 (± 1.07)	03.12 (± 1.30)	12.29 (± 1.68)	22.51 (± 1.56)	02.42 (± 0.21)	03.07 (± 0.55)
CLIP (Radford et al., 2021) (*zero shot*)	00.00 (± 0.01)	00.05 (± 0.04)	00.54 (± 0.58)	01.10 (± 1.02)	00.22 (± 0.16)	00.23 (± 0.16)
ViLT (Kim et al., 2021)	00.70 (± 0.10)	04.97 (± 0.46)	15.66 (± 1.06)	22.73 (± 1.75)	01.79 (± 0.22)	02.09 (± 0.25)
MIMIC (Luo et al., 2023)	27.50 (± 2.41)	60.95 (± 2.16)	77.11 (± 0.66)	82.40 (± 0.32)	47.24 (± 3.43)	66.29 (± 3.85)
Mulang et al. (2020)	12.33 (± 0.89)	43.63 (± 0.84)	64.21 (± 1.40)	71.89 (± 1.46)	38.98 (± 1.45)	68.03 (± 1.76)
KGMEL (Kim et al., 2025)	18.61 (± 0.52)	42.69 (± 0.78)	59.98 (± 0.67)	66.62 (± 0.53)	31.78 (± 0.67)	48.11 (± 0.98)
UniMEL (Liu et al., 1909)	11.52 (± 0.45)	16.20 (± 0.68)	26.23 (± 1.47)	31.74 (± 1.52)	21.19 (± 0.76)	24.63 (± 0.87)
GEMEL (Shi et al., 2024)	13.57 (± 0.10)	–	–	–	–	–
KARAMEL (*Ours*)	**29.67 (± 1.91)**†	**62.87 (± 2.19)**†	**77.66 (± 0.64)**†	**83.04 (± 0.23)**	**50.56 (± 3.18)**†	**69.96 (± 3.70)**†
MELArt (Sierra-Múnera et al., 2024)						
Model	R@1	R@10	R@50	R@100	MAP	MRR
BERT (Devlin et al., 2019)	03.84 (± 4.30)	18.54 (± 5.41)	41.21 (± 3.53)	55.20 (± 5.12)	11.16 (± 5.11)	14.60 (± 7.33)
CLIP (Radford et al., 2021)	12.55 (± 4.22)	31.67 (± 3.84)	50.93 (± 3.69)	60.44 (± 4.43)	02.06 (± 0.69)	02.03 (± 0.69)
CLIP (Radford et al., 2021) (*zero shot*)	00.00 (± 0.00)	00.05 (± 0.11)	02.40 (± 1.58)	05.16 (± 1.87)	00.75 (± 0.31)	00.75 (± 0.31)
ViLT (Kim et al., 2021)	01.52 (± 1.14)	07.95 (± 4.94)	26.19 (± 13.20)	43.90 (± 11.54)	02.48 (± 0.69)	02.84 (± 1.07)
MIMIC (Luo et al., 2023)	50.46 (± 2.47)	82.76 (± 1.08)	**91.27 (± 1.07)**	**95.04 (± 0.85)**	70.07 (± 2.45)	81.79 (± 1.86)
Mulang et al. (2020)	21.30 (± 0.84)	55.50 (± 0.79)	60.80 (± 1.15)	60.80 (± 1.15)	49.85 (± 1.67)	54.47 (± 1.43)
KGMEL (Kim et al., 2025)	39.83 (± 0.85)	71.63 (± 1.66)	82.94 (± 0.86)	87.77 (± 1.08)	57.96 (± 0.31)	67.15 (± 1.21)
UniMEL (Liu et al., 1909)	28.10 (± 1.85)	37.09 (± 3.87)	60.79 (± 2.82)	73.21 (± 1.89)	36.27 (± 2.04)	43.24 (± 2.89)
GEMEL (Shi et al., 2024)	36.38 (± 1.37)	–	–	–	–	–
KARAMEL (*ours*)	**51.67 (± 1.79)**	**82.96 (± 1.38)**	91.05 (± 1.55)	94.26 (± 0.65)	**71.39 (± 1.94)**	**82.12 (± 1.91)**

Each cell reports the mean and standard deviation (in parentheses). All metrics are expressed in the range [0, 100]. The best performance per metric is shown in bold. Statistical significance over the second best is marked by†

implicit entity references. This highlights the need for a dedicated, context-aware solution to achieve better performance.

Lastly, compared to MIMIC, KARAMEL yields consistent improvements across all metrics on the *WikiMuSA* dataset, with a particularly clear gain in MAP and a lower variance

across seeds, indicating more stable ranking behavior. Beyond aggregate performance, the benefit of structured context becomes most apparent in discovery-oriented cases where evidence is incomplete—e.g., when entities are not explicitly mentioned in the text or when depictions are more abstract—supporting the role of graph-based contextual reasoning for disambiguation. This also shows that encoding entities using a KG, in this case $\mathcal{WikiMuSA}$, is beneficial for exploiting correlations among entities, detected by graph convolutions. For example, knowing that a candidate entity is depicted in specific artworks (i.e., the *depicts* relationship—first-order information) provides insight into the artistic expressiveness associated with that entity. Additionally, leveraging metadata about those artworks (e.g., their *styles*, *genres*, etc.—second-order information) enriches the learned entity representation by highlighting the variations in how an entity may be portrayed depending on these factors. Finally, recognizing that multiple entities, for example, share the same social position (i.e., the *position held* relationship—first-order information) enables the model to exploit these correlations, resulting in more robust representations. In contrast, MIMIC relies solely on vision-text information and is unable to exploit the aforementioned features, resulting in a pure independent-entity encoding strategy that performs worse. To verify that the observed improvements on MIMIC are not attributable to random variation, we conducted statistical significance testing using both a one-sided Wilcoxon signed-rank test and a paired t-test. We compared the two models by computing each metric at the level of individual mentions and forming paired observations accordingly. As a representative run, we selected the seed with the median MAP among all seeded runs for each model. A difference was considered statistically significant only if both tests produced p-values below 0.05. The results indicate that the improvements over MIMIC are statistically significant for all metrics except Recall@100.

When shifting our focus to the MELArt dataset, the evaluated models generally maintain their relative rankings. The KG-based and LLM-based approaches continue to trail behind the multimodal solutions, with KGMEL achieving the highest scores among them, followed by Mulang et al. and UniMEL. At the top of the scale, KARAMEL and MIMIC remain the best-performing models; however, unlike our observations on $\mathcal{W}$ *iki* $\mathcal{M}$ *u* $\mathcal{SA}$, the performance gap between them is considerably smaller on MELArt. In fact, after conducting the same statistical significance testing between KARAMEL and MIMIC, we cannot reject the null hypothesis. We attribute this to key differences between the datasets: although both are in the same domain, MELArt is considerably smaller and primarily consists of artworks where entities are explicitly mentioned in the text. In contrast, $\mathcal{W}$ *iki* $\mathcal{M}$ *u* $\mathcal{SA}$ contains a larger proportion of implicit references, where entities are not directly stated but must be inferred from contextual and relational signals. In these cases, the graph modality becomes more informative, leading to clearer and statistically significant gains.

5.3.2 Feature-Wise Statistical Analysis

To address RQ2 and validate the effectiveness of the ontological information in $\mathcal{WikiMuSA}$, Fig. 4 presents a detailed statistical significance analysis comparing the rankings produced by KARAMEL and MIMIC. We focus here on MIMIC because its aggregated ranking performance is closest to KARAMEL, and KARAMEL's image-text cross-encoder is based on MIMIC, allowing us to evaluate the impact of including the knowledge graph in the ranking. Specifically, we considered all (*artwork*, *depicted entity*) test pairs and computed

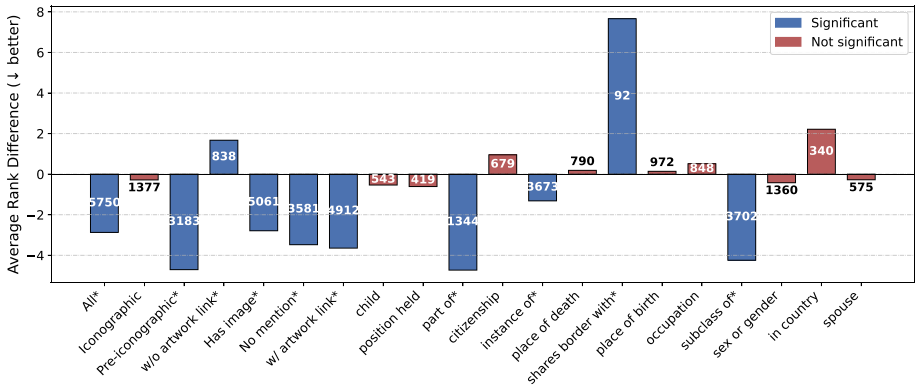


Fig. 4 Average rank difference of KARAMEL on MIMIC (Luo et al., 2023). Bars below the x-axis indicate improved ranking of KARAMEL. Numbers within each bar denote the number of (*artwork*, *depicted entity*) pairs supporting the corresponding feature. Blue bars denote statistically significant effects, whereas red bars indicate non-significant effects.

the rank difference as the rank predicted by KARAMEL minus that predicted by MIMIC. Throughout our analysis, a negative value indicates that KARAMEL ranked the correct entity higher than MIMIC (i.e., closer to the top), reflecting improved performance, whereas a positive value indicates worse performance. As a base case, if the ground-truth entity did not appear in the top 100 results for either model, we assigned it a default rank of 101. We performed a Wilcoxon signed-rank test and a paired t-test statistical significance test on the ranks grouped for each contextual characteristic.

We further enriched our analysis by reporting quantitative results based on both ontological and non-ontological characteristics of artworks and depicted entities. First, using IICONGRAPH (Sartini, 2024), we categorized each depiction as either iconographic or pre-iconographic. Next, we examined whether the availability of images associated with the depicted entities influenced ranking quality. We also analyzed whether entities linked to artworks in the training graph (*with artwork link*) exhibited different ranking behavior compared to those without such links (*without artwork link*), under the condition that all associated artworks appeared only in the validation or test sets. Additionally, we investigated whether the input text explicitly mentioned the depicted entity. To this end, we employed a rule-based phrase-matching algorithm to identify English labels. For instance, in our motivating example shown in Fig. 1, the entity “ruins” is not explicitly mentioned in the text. Finally, we assessed the impact of specific knowledge graph predicates by verifying whether the entity appeared as a subject in at least one triple containing a given predicate within the subgraph, thereby enabling us to evaluate the relevance of these predicates to the entity representations.

As shown in Fig. 4, KARAMEL significantly outperforms MIMIC in several cases, corroborating the findings of RQ1. Notably, substantial improvements are observed when the entity is not explicitly mentioned in the text. This suggests that contextual embeddings derived from the knowledge graph enhance the expressiveness of entity representations beyond what is available in the textual description alone. Moreover, performance gains are particularly pronounced for pre-iconographic entities, which are typically more abstract and thus benefit more from ontological relations such as *subclass of*. Similarly, entities associ-

ated with predicates such as *part of*, *subclass of*, or *instance of* tend to yield better results, as these relations provide valuable structural and semantic context. Furthermore, entities linked to artworks in the graph (*w/artwork link*) exhibit improved rankings, confirming that modeling how an entity is visually depicted through entity–artwork links enriches its representation. Conversely, for entities without artwork links (*w/o artwork link*), MIMIC outperforms KARAMEL, further confirming that integrating exemplifying information to depicted entities leads to improved rankings.

Additionally, to qualitatively illustrate the differences in model behavior, we present an example from the test set in Table 3. In this case, KARAMEL correctly ranks both ground-truth entities—*Polyphemus* and *Odysseus*—at the top-4, while MIMIC ranks only one of them within the top-4. This example also shows that both models retrieve semantically plausible candidates; however, KARAMEL's scores align more closely with the entities considered most significant according to Wikidata's annotations. Furthermore, KARAMEL presents a semantically cohesive ranking, suggesting that the KG entity encoding produces more semantically valuable information, thus improving the model's expressiveness.

5.3.3 Complexity and Scalability Analysis

As detailed in Sect. 4.2, KARAMEL incorporates a graph modality alongside vision and text. As shown in Table 4, this architectural choice naturally increases the number of learnable parameters compared to non-KG multimodal baselines such as MIMIC (260.14M vs. 152.72M), primarily due to the message-passing operations required to extract the contextual entity representation matrix K. While this results in higher offline costs in terms of training time per batch (0.34s vs. 0.15s for MIMIC) and entity embedding extraction time (0.17s vs. 0.09s), online throughput remains essentially unchanged, with comparable validation time per batch (0.87s vs. 0.85s).

Table 3 Top-4 ranking comparison for "*Odysseus and Polyphemus*" (**Q20537891**), a painting by Arnold Böcklin (1896), between KARAMEL and MIMIC (Luo et al., 2023).

Image & Text		MIMIC	KARAMEL
 *The artwork "Odysseus and Polyphemus" represents a pivotal moment from Homer's epic poem, where Odysseus taunts the one-eyed Cyclops Polyphemus as he prepares to hurl a rock at the escaping boat of Odysseus and his crew.*	1	 **Odysseus**. Greek king of Ithaca.	 **Odysseus**. Greek king of Ithaca.
	2	 **oar**. water-borne propulsion	 **oar**. water-borne propulsion
	3	 **fisher**. someone who captures fish.	 **cyclops**. giant in Greek mythology.
	4	 **barque**. type of sailing vessel.	 **Polyphemus**. giant son of Poseidon.
Ground Truth:		 **Polyphemus**. giant son of Poseidon.	 **Odysseus**. Greek king of Ithaca.

Ground truth entities are highlighted

Table 4 Computational complexity comparison between KARAMEL and the experimented baselines.

Model	# Parameters (M)	Train		Validation		Entity Extraction	
		Duration (min)	Batch (s)	Duration (min)	Batch (s)	Duration (min)	Batch (s)
BERT (Devlin et al., 2019)	110.27	1.80	0.15 (± 0.13)	0.17	0.13 (± 0.03)	0.11	0.03 (± 0.01)
CLIP (Radford et al., 2021)	151.80	2.52	0.20 (± 0.14)	0.34	0.13 (± 0.03)	0.27	0.07 (± 0.00)
ViLT (Kim et al., 2021)	112.38	3.10	0.24 (± 0.28)	0.44	0.16 (± 0.04)	0.36	0.10 (± 0.01)
MIMIC (Luo et al., 2023)	152.72	1.97	0.15 (± 0.08)	0.91	0.85 (± 0.06)	0.34	0.09 (± 0.02)
Mulang et al. (2020)	124.65	13.92	2.77 (± 0.12)	1468.33	2.94e4 (± 6.73e3)	–	–
KGMEL (Kim et al., 2025)	154.69	24.28	1.81 (± 0.16)	3.58	1.17 (± 0.21)	2.94	0.84 (± 0.11)
UniMEL (Liu et al., 1909)	20.97	22.29	0.39 (± 0.1)	0.06	0.11 (± 5e-4)	16.55	0.08 (± 8e-4)
GEMEL (Shi et al., 2024)	16.79	117.00	0.65 (± 0.04)	31.00	1.38 (± 0.10)	–	–
KARAMEL (*ours*)	260.14	4.21	0.34 (± 0.06)	0.42	0.87 (± 0.05)	0.58	0.17 (± 0.01)

We report the number of learnable parameters (in millions), total training and validation durations, and entity embedding extraction times. Batch columns indicate the average processing time per batch (mean ± standard deviation) in seconds.

Crucially, when evaluated against other KG-aware and generative baselines, KARAMEL demonstrates a highly favorable computational profile. Although LLM-based models like UniMEL and GEMEL report fewer learnable parameters (20.97M and 16.79M, respectively), these figures represent only their trainable parameter-efficient fine-tuning modules; their massive, frozen LLM backbones inevitably dictate a substantially larger memory footprint.[2] Furthermore, KARAMEL achieves remarkable time efficiency compared to these advanced paradigms. KG-based competitors such as KGMEL and Mulang et al. exhibit severe computational bottlenecks, requiring 24.28m and 13.92m for training, respectively, with significantly higher per-batch processing costs. Similarly, generative LLM approaches suffer from prolonged training durations (e.g., 117.00m for GEMEL) and expensive validation phases due to autoregressive decoding. Consequently, KARAMEL effectively circumvents the prohibitive computational costs typically associated with KG-aware and generative state-of-the-art approaches, establishing a clear and optimal trade-off between representational capacity and computational tractability.

Additionally, in practical settings the number of candidate entities can grow substantially, making exhaustive mention–entity comparison computationally prohibitive. To address this challenge, we investigated the effectiveness of candidate pre-filtering strategies that reduce the search space prior to mention–entity matching, thereby improving efficiency while preserving retrieval performance. Such pre-filtering is particularly important in matching-based MEL approaches, where computational costs scale directly with the number of candidate

[2] Following the authors' original configurations and to account for memory limits, both UniMEL and GEMEL were trained using a batch size of one

entities that must be compared against each mention. Specifically, we leverage the multimodal encoder (see Sect. 4.2.1) to extract textual and visual representations for both mentions and entities, focusing on the global textual embeddings (i.e., $\mathrm{t}^G_{M_i}$, $\mathrm{t}^G_{E_j}$) and the global visual embeddings (i.e., $\mathrm{V}^G_{M_i}$, $\mathrm{V}^G_{E_j}$). We then construct two independent FAISS (Johnson et al., 2019) indices, one for textual entity embeddings and own for visual entity embeddings. Given a mention, nearest-neighbor retrieval is performed independently in both embedding spaces, yielding modality-specific similarity scores:

$$s^t_{i,j} = \mathrm{sim}(\mathrm{t}^G_{M_i}, \mathrm{t}^G_{E_j}),\ s^v_{i,j} = \mathrm{sim}(\mathrm{V}^G_{M_i}, \mathrm{V}^G_{E_j}), \tag{13}$$

where $\mathrm{sim}(\cdot,\cdot)$ denotes the similarity function employed by the FAISS retriever (in our case, the cosine similarity score). The final retrieval score is obtained by averaging the modality-specific scores:

$$s_{i,j} = \frac{\left(s^t_{i,j} + s^v_{i,j}\right)}{2}. \tag{14}$$

Entities are subsequently ranked to $s_{i,j}$, and only the top-k candidates are retained for the downstream mention-entity matching stage. Figure 5 reports a scalability analysis under different candidate set sizes, obtained by retaining the top-k entities after the pre-filtering stage, where $k \in \{10, 50, 100, 500, 1000\}$. We compare KARAMEL against MIMIC—our most scalable and competitive multimodal baseline—to assess how varying the number of candidate entities affects both retrieval effectiveness and computational efficiency. Specifically, KARAMEL consistently outperforms MIMIC across all candidate set sizes in terms of Recall@10 and MAP, while exhibiting a comparable growth in computational cost as the number of candidates increases. This suggests that the proposed approach scales well with the size of the candidate space and can naturally benefit from a query-specific candidate generation stage when operating over large-scale knowledge bases.

5.4 Ablation Studies

To address RQ3, we conducted different ablation studies on KARAMEL, including the impact of ontological information richness in our proposed framework, the feature encoding, our proposed loss, and the quality of the translation-summarization step.

5.4.1 Graph Encoding Variations

Firstly, we evaluated the impactfulness of ontological information within *WikiMuSA*, identifying two graph versions, namely the *full* and the *ablated* graph. The former encodes rich, heterogeneous contextual relations around each *Depicted Object*, capturing how depicted entities are factually expressed and situated within broader semantic and ontological structures. These relations connect entities to types, attributes, and other contextual elements that shape their meaning beyond their immediate visual appearance. In contrast, the ablated graph retains only a minimal shared ontological link—(*Depicted Object, instance of, Class*)—removing most other contextual edges. In addition, we made a complementary ablation on raw node features encoding. Specifically, we compared three encoding strate-

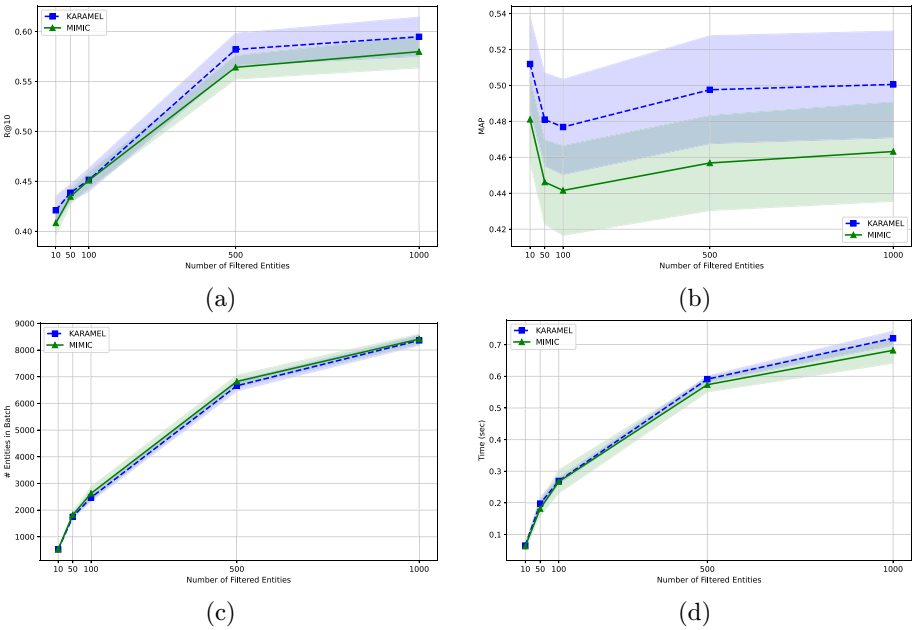


Fig. 5 Scalability analysis with varying numbers of pre-filtered candidate entities $k \in \{10, 50, 100, 500, 1000\}$. Comparison between KARAMEL and MIMIC in terms of **a** R@10, **b** MAP, **c** entity extraction time, and **d** mention–entity matching time. Shaded regions indicate standard deviation across runs

gies. In the textual setting, each node is described solely by a CLIP text embedding derived from its Wikidata label and description, prioritizing linguistic semantics. In the multimodal setting, text and image embeddings are combined via summation so that entities with available visuals can leverage both modalities (with zero-padding used when images are missing). Finally, the hybrid configuration uses visual features for artwork nodes while encoding all other entities textually, reflecting the intuition that artworks are best recognized visually whereas contextual entities (e.g., people, places, concepts) are better characterized through language.

Across all encoding strategies, the results in Table 5 show a clear and consistent drop in performance when moving from the full graph to the ablated graph. This indicates that rich ontological and contextual relations play a crucial role in enabling more expressive and discriminative representations of depicted entities, helping the model capture how they are factually grounded in context rather than treated as isolated visual instances.

With respect to node feature encodings, the hybrid approach yields the best overall ranking performance in terms of MAP and MRR on the full graph, suggesting that a domain-aware combination of visual and textual signals provides the most balanced and effective representation. Interestingly, the purely textual model achieves the highest Recall@10, demonstrating that textual embeddings alone remain highly competitive, particularly in early retrieval settings. This is practically meaningful since end users typically inspect only the top-k results, making Recall@10 a valuable indicator of real-world retrieval utility.

Table 5 Ablation study on the effect of both contextual factors (i.e., Full Graph) and node feature encoding variations (i.e., Textual, Hybrid, and Multimodal).

Full Graph	Encoding	R@10	MAP	MRR
✗	Multimodal	60.56 (± 2.02)	47.56 (± 2.63)	66.44 (± 2.60)
✗	Hybrid	61.83 (± 1.49)	48.82 (± 2.12)	67.97 (± 2.65)
✗	Textual	61.26 (± 1.73)	47.95 (± 2.11)	66.86 (± 2.26)
✓	Multimodal	60.89 (± 0.66)	47.25 (± 0.80)	66.38 (± 0.95)
✓	Hybrid	60.66 (± 0.35)	47.68 (± 0.37)	66.95 (± 0.60)
✓	Textual	**62.87 (± 2.19)**	**50.56 (± 3.18)**	**69.96 (± 3.70)**

Numbers in bold indicate the best results.

Table 6 Ablation study on loss components impact.

Loss Component				R@1	R@10	R@50	R@100	MAP	MRR
$\mathcal{L}_C$	$\mathcal{L}_T$	$\mathcal{L}_V$	$\mathcal{L}_{VT}$						
✓	✗	✗	✗	17.79 (± 1.08)	44.42 (± 1.30)	62.61 (± 0.72)	70.96 (± 0.79)	33.51 (± 1.47)	49.77 (± 1.68)
✓	✓	✗	✗	28.24 (± 1.63)	61.53 (± 1.96)	77.01 (± 0.90)	82.57 (± 0.40)	48.45 (± 2.90)	67.71 (± 2.86)
✓	✗	✓	✗	16.90 (± 1.81)	44.23 (± 2.08)	62.52 (± 1.07)	70.44 (± 0.75)	32.23 (± 2.73)	48.55 (± 4.00)
✓	✗	✗	✓	24.90 (± 0.40)	55.66 (± 1.64)	72.80 (± 0.77)	79.30 (± 0.40)	43.18 (± 1.17)	61.92 (± 0.86)
✓	✓	✓	✗	27.62 (± 2.02)	60.86 (± 2.04)	76.71 (± 0.95)	82.28 (± 0.58)	47.22 (± 3.08)	66.59 (± 3.33)
✓	✓	✗	✓	27.89 (± 1.98)	60.11 (± 2.61)	75.78 (± 1.20)	81.50 (± 0.78)	47.25 (± 3.25)	66.58 (± 3.31)
✓	✗	✓	✓	24.58 (± 1.68)	57.02 (± 2.83)	74.15 (± 1.31)	80.20 (± 1.01)	43.53 (± 3.12)	62.19 (± 3.46)
✗	✓	✓	✓	27.50 (± 2.41)	60.95 (± 2.16)	77.11 (± 0.66)	82.40 (± 0.32)	47.24 (± 3.43)	66.29 (± 3.85)
✓	✓	✓	✓	**29.67 (± 1.91)**	**62.87 (± 2.19)**	**77.66 (± 0.64)**	**83.04 (± 0.23)**	**50.56 (± 3.18)**	**69.96 (± 3.70)**

Numbers in bold indicate the best results.

5.4.2 Loss Components

We analyze the contribution of each component of the proposed objective function (see Sect. 4.2.2) through a systematic ablation study. Starting from a minimal configuration, we progressively activate each component and evaluate all combinations to assess their individual and joint impact on ranking performance. The results are reported in Table 6. Using only the contextual loss $\mathcal{L}_C$ yields clearly suboptimal performance across all metrics. This behavior is expected, as $\mathcal{L}_C$ alone does not enforce explicit single-modality constraints, limiting its ability to learn discriminative representations for the MEL task. In contrast, incorporating text-based supervision leads to substantial improvements. In particular, the configuration combining $\mathcal{L}_T$ and $\mathcal{L}_C$ already achieves strong performance, surpassing the

MIMIC baseline (i.e., $\mathcal{L}_C$, $\mathcal{L}V$, $\mathcal{L}_{VT}$) and demonstrating that the combination of textual and contextual information is highly effective. However, the best performance is obtained by KARAMEL, indicating that contextual supervision, when combined with both single-modality alignment and cross-modal objectives, provides complementary information that yields more expressive representations. This highlights the importance of a fully integrated multimodal learning formulation for MEL.

5.4.3 Text Encoder

Embedding textual information is fundamental in our architecture, as we rely on text representations for multiple components of the model. In particular, we encode both mention and entity textual descriptions, and we also use text embeddings as raw node features within the graph, making the choice of text encoder a key design factor of the overall framework. To this end, we evaluated the impact of the text encoder size and pretraining on the proposed methodology. Specifically, we select four representative text encoders—BGE (Chen et al., 2024), DeBERTa (He et al., 2020), ModernBERT (Warner et al., 2025), and RoBERTa (Liu et al., 2019)—chosen to cover a spectrum of increasing model capacity and representational expressiveness, while also including encoders specifically designed for retrieval-oriented tasks. In order to ensure a fair and controlled evaluation, we fully replace the original text encoder (BERT pretrained with CLIP) with each of the selected models. This substitution is applied consistently across all components of the pipeline, including mention/entity feature extraction as well as raw node feature extraction. The results are reported in Table 7. Overall, performance varies across encoder choices, reflecting their differing pretraining objectives and representational strengths. DeBERTa yields comparatively weaker results among the evaluated models, while RoBERTa and ModernBERT provide more competitive and stable performance across all metrics. BGE achieves the strongest overall results, which is consistent with its retrieval-oriented pretraining and demonstrated effectiveness in dense retrieval settings. Across all configurations, however, the best performance is obtained when using the original CLIP-pretrained text encoder. This indicates that, despite the strong performance of recent retrieval-optimized and large-scale language encoders, the multimodal pretraining used in CLIP provides a more effective initialization for the considered task. These results suggest that cross-modal alignment plays a central role in the overall performance, and that it is more influential than increasing text encoder capacity alone, even when using state-of-the-art retrieval-focused encoders (see Fig. 6).

5.4.4 Translation and Summarization

Although we conducted a qualitative evaluation of the *WikiMuSA* summaries on a sample of the dataset, confirming their semantic alignment with the original Wikipedia articles, we further performed an ablation study to assess both the impact of using raw Wikipedia text and the effectiveness of the translation–summarization pipeline. Additionally, Fig. 7 presents a qualitative example illustrating the transformation of a Wikipedia article throughout the preprocessing pipeline, from the original multilingual version to its English translation and final summarized representation. To facilitate comparison, we highlight the named entities in each version, emphasizing how the most relevant concepts are preserved despite the substantial reduction in document length.

Table 7 Performance comparison of KARAMEL and MIMIC under different text encoder backbones (BGE, DeBERTa, ModernBERT, and RoBERTa).

Text Encoder	Model	R@1	R@10	R@50	R@100	MAP	MRR
BGE (Chen et al., 2024)	KARAMEL	25.56 (± 0.93)	57.19 (± 1.47)	74.68 (± 0.35)	80.66 (± 0.34)	43.75 (± 1.39)	62.94 (± 1.62)
	MIMIC (Luo et al., 2023)	25.03 (± 0.96)	56.02 (± 2.23)	73.59 (± 1.78)	79.92 (± 0.94)	42.46 (± 2.12)	61.63 (± 2.65)
DeBERTa (He et al., 2020)	KARAMEL	13.90 (± 2.28)	41.50 (± 2.80)	61.81 (± 2.03)	70.43 (± 1.22)	28.59 (± 3.24)	43.78 (± 4.71)
	MIMIC (Luo et al., 2023)	14.06 (± 1.00)	41.57 (± 1.61)	61.54 (± 1.70)	70.18 (± 1.27)	28.42 (± 1.43)	42.66 (± 1.75)
ModernBERT (Warner et al., 2025)	KARAMEL	25.15 (± 1.13)	56.97 (± 2.10)	74.01 (± 1.42)	80.23 (± 1.00)	43.72 (± 2.04)	62.58 (± 2.38)
	MIMIC (Luo et al., 2023)	24.19 (± 0.88)	56.66 (± 1.44)	73.51 (± 0.36)	79.67 (± 0.49)	42.64 (± 1.66)	61.02 (± 2.16)
RoBERTa (Liu et al., 2019)	KARAMEL	24.09 (± 1.42)	55.32 (± 1.37)	73.15 (± 0.94)	79.43 (± 0.35)	41.79 (± 1.83)	59.95 (± 2.40)
	MIMIC (Luo et al., 2023)	23.20 (± 1.44)	54.99 (± 1.74)	72.86 (± 1.19)	78.85 (± 0.90)	40.62 (± 2.21)	58.30 (± 2.73)
BERT (Devlin et al., 2019) (*ours*)	KARAMEL	**29.67 (± 1.91)**	**62.87 (± 2.19)**	**77.66 (± 0.64)**	**83.04 (± 0.23)**	**50.56 (± 3.18)**	**69.96 (± 3.70)**
	MIMIC (Luo et al., 2023)	27.50 (± 2.41)	60.95 (± 2.16)	77.11 (± 0.66)	82.40 (± 0.32)	47.24 (± 3.43)	66.29 (± 3.85)

All metrics are expressed in the range [0, 100]. The best performance per metric is shown in bold.



Fig. 6 Performance comparison between KARAMEL and MIMIC using raw (full) articles versus summarized articles on the *WikiMuSA* dataset.

Figure 6 reports the retrieval performance of KARAMEL and MIMIC when the raw, full-length Wikipedia articles are used instead of their summarized counterparts. First, we observe that the overall trend across metrics remains consistent: KARAMEL outperforms MIMIC regardless of the textual input format, which suggests that our context-aware graph representations provide effective and expressive information for this task.

Alongside the retrieval metrics, Table 8 quantifies the fidelity of the preprocessing pipeline through several complementary metrics computed for three transformations: Original-to-Translated (O→T), Translated-to-Summarized (T→S), and Original-to-Summarized (O→S). These include semantic similarity measures such as BERTScore and MoverScore, which evaluate contextual and embedding-based alignment between texts; lexical overlap metrics such as ROUGE-L (based on longest common subsequence overlap), chrF (character-level n-gram overlap), and METEOR (alignment-based matching with stemming and synonymy); and entity-level measures including Entity Precision, Recall, F1, and Keyword



Fig. 7 Qualitative example of the translation and summarization pipeline used to construct the textual mentions in *WikiMuSA*. Starting from the original multilingual Wikipedia article (Spanish in this example), the article is first translated into English and subsequently summarized to produce a concise description. Named entities are highlighted in each stage to illustrate key aspects included in each textual description.

Recall, which assess the preservation of salient named entities and content words. We also report Compression Ratio, capturing the degree of textual reduction, and Readability, which reflects the fluency and ease of understanding of the generated summaries.

The translation stage exhibits very high semantic preservation. Both BERTScore and MoverScore indicate that the translated articles remain almost semantically equivalent to the original multilingual Wikipedia pages, suggesting that the translation process retains the

Table 8 Quantitative evaluation of the translation and summarization pipeline used to construct the *WikiMuSA* textual mentions

Metric	O→T	T→S	O→S
BERTScore	97.62	98.18	97.64
MoverScore	98.52	97.87	97.24
ROUGE-L	51.48	19.45	10.77
chrF	48.29	11.39	5.72
METEOR	48.52	11.89	5.92
Entity F1	63.44	26.83	18.35
Entity Precision	63.45	47.60	32.30
Entity Recall	64.17	20.36	13.91
Keyword Recall	51.16	20.04	10.02
Compression Ratio	–	30.77	31.17
Readability	–	4.95	3.56

Results are reported for three transformations: Original-to-Translated (O→T), Translated-to-Summarized (T→S), and Original-to-Summarized (O→S).

original semantic content with minimal information loss. Moderate lexical-overlap scores (e.g., ROUGE-L and METEOR) are expected, as translation naturally alters the surface form while preserving meaning. Conversely, the summarization stage performs a much stronger transformation. Although lexical-overlap metrics decrease substantially, semantic similarity remains remarkably high. This apparent discrepancy reflects the abstractive nature of the summarization process, which preserves the core semantic content while extensively rephrasing and compressing the original text.

6 Conclusion

In this paper, we introduced KARAMEL, a context-aware framework for multimodal entity linking in the arts that integrates vision and language with structured relational knowledge. By combining a vision–language backbone with a heterogeneous graph encoder, KARAMEL leverages contextual dependencies to improve entity ranking and disambiguation in a challenging discovery setting, where relevant entities may be implicit and textual mention spans are not pre-defined. Experiments on both the proposed *WikiMuSA* dataset and the MELArt dataset show consistent gains over strong baselines, highlighting the benefit of domain-relevant knowledge for robust multimodal linking.

A key strength of KARAMEL lies in graph-driven contextual reasoning: entities are represented not only by their local multimodal evidence but also by their positions within a heterogeneous relational structure. This enables the model to capture correlations and dependencies that are difficult to infer from vision or text alone, particularly for abstract or implicitly referenced entities. While our study focuses on cultural heritage, the proposed design may also be extended to other domains.

Several directions for future work follow naturally. First, automatic and adaptive subgraph extraction could reduce reliance on manually curated predicates while improving scalability on large linked open data sources. Second, integrating explainable AI mechanisms—e.g., evidence attribution over graph neighborhoods and cross-modal interactions—could increase transparency and trustworthiness for downstream users. Finally, a curated manual annotation of article translation and summarization could further enhance the quality of the proposed *WikiMuSA*.

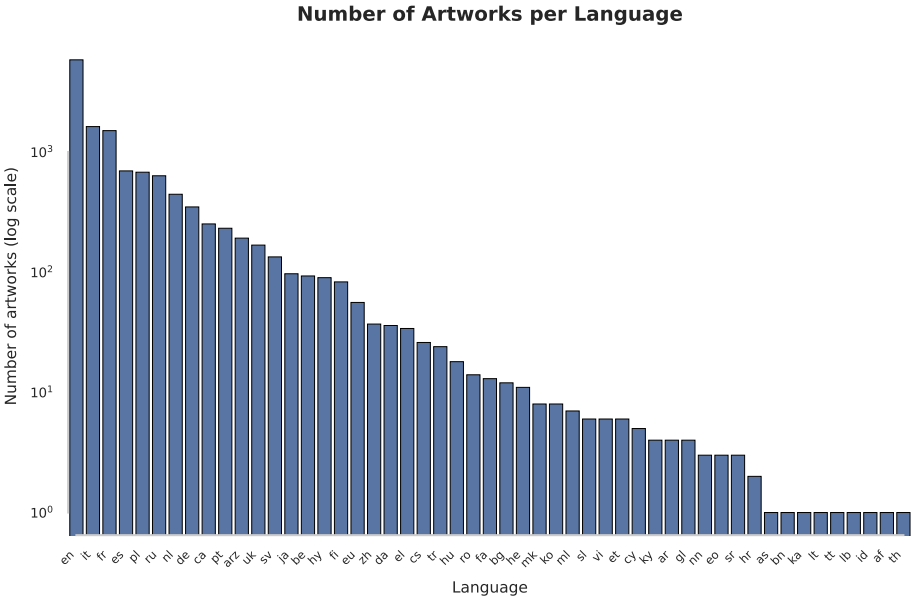


Fig. 8 Distribution of artwork Wikipedia articles (*x*-axis) by original language (*y*-axis)

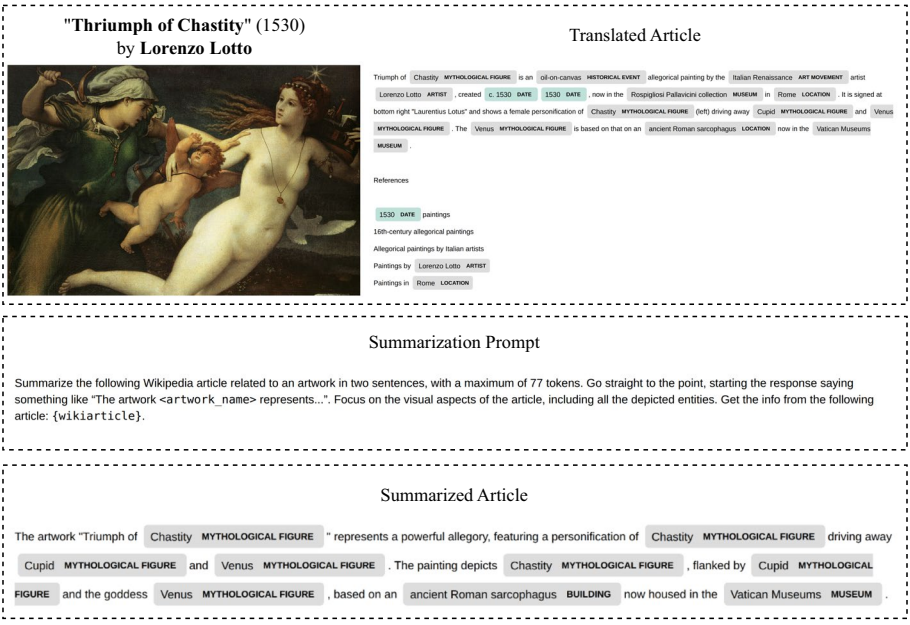


Fig. 9 Example of summarization used to compress textual mentions in *WikiMuSA*. Starting from the english version of the Wikipedia article, we ask LLaMA to summarize it using a specific instruction prompt. Named entities are highlighted to illustrate key aspects included in both input and resulting textual description

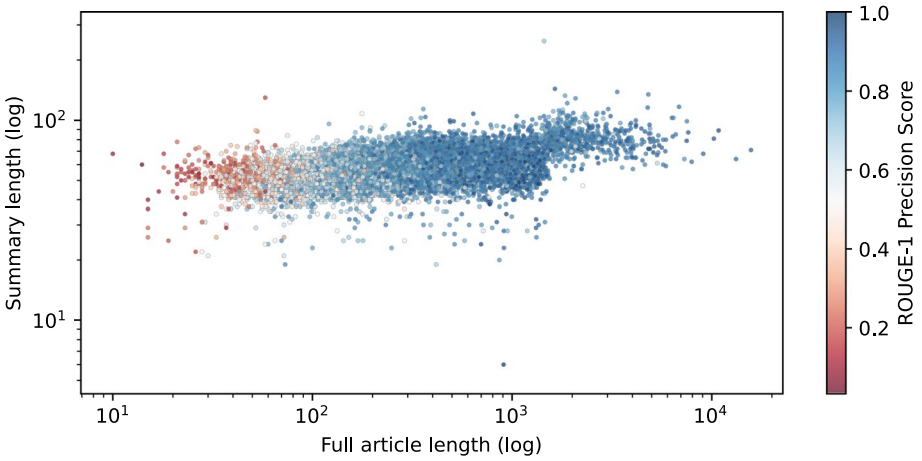


Fig. 10 Length in words of the original articles and their summaries. The color represents the ROUGE-1 precision for each article-summary pair

Translation and Summarization Analysis

We extracted Wikipedia articles from the snapshot updated at 2023/11/01[3], matching Wikipedia titles and languages from Wikidata. When the title was not found in the snapshot, we expanded the search using the redirect query https://<LANG>.wikipedia.org/w/api.php?action=query&prop=redirects, since some articles may have changed their titles during the time span between Wikidata's and Wikipedia's snapshots. For each artwork that has no English Wikipedia article, we translated one article in a different language following these steps:

1. Filter articles and keep the languages supported by NLLB
2. Choose the longest article
3. Split the article into paragraphs;
4. Translate each paragraph independently;
5. Concatenate the translated paragraphs.

Figure 8 shows the proportion of languages from the selected articles in the dataset.

We balanced the length and level of detail in the Wikipedia articles summarizing them using Llama 3 (Grattafiori et al., 2024). For each article, we used a fixed instruction prompt to extract the summary, as illustrated in Fig. 9.

The result is a more balanced article length, which can be seen in Fig. 10. We see that the summary length is concentrated between 30 and 100 word texts, while the original articles ranged between 20 and more than 1000 words.

Acknowledgements We acknowledge the CINECA award under the ISCRA initiative, which gave us access to computing resources and support. We thank the Information and Data Engineering Research School from HPI for contributing to funding this research.

[3] https://huggingface.co/datasets/wikimedia/wikipedia

Author Contributions Raffaele Scaringi: Conceptualization, Methodology, Validation, Data Curation, Writing - Original Draft. Alejandro Sierra-Múnera: Conceptualization, Methodology, Validation, Data Curation, Writing - Original Draft. Gennaro Vessio: Conceptualization, Writing - Original Draft, Supervision. Giovanna Castellano: Conceptualization, Writing - Review & Editing, Supervision. Ralf Krestel: Conceptualization, Writing - Review & Editing, Supervision

Data Availability The dataset is publicly available on Zenodo (https://zenodo.org/records/23019880)

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Authors and Affiliations

Raffaele Scaringi[1] · Alejandro Sierra-Múnera[2] · Gennaro Vessio[1] · Giovanna Castellano[1] · Ralf Krestel[3,4]

✉ Raffaele Scaringi
raffaele.scaringi@uniba.it

[1] Department of Computer Science, University of Bari Aldo Moro, 70125 Bari, Italy

[2] Hasso Plattner Institute, University of Potsdam, 14482 Potsdam, Germany

[3] University of Kiel, 24118 Kiel, Germany

[4] ZBW - Leibniz Information Centre for Economics, 24105 Kiel, Germany